Ericsson PE Node (WHI-PE-14) Optical Transport Failure Resolution SOP

Node: WHI-PE-14 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_9755

Description

This SOP addresses a warning-level Optical Transport Failure on Ericsson PE node WHI-PE-14 at the Whitefield East POP, triggered by Alarm Code ALARM_9755. The failure domain is primarily attributed to degradation of SFP/XFP components, fiber cuts, or Rx/Tx power loss within the 10G-ER-LD optical transport module. The 10G-ER-LD module employs a 10-Gbps, 1550-nm wavelength, optical transport system that relies on the SONET/SDH framing protocol for error detection and correction. The underlying hardware mechanics involve optoelectronic conversion of optical signals into electrical signals, where a SFP/XFP module's degradation or fiber cuts can impede this conversion process, resulting in Rx/Tx power loss and, consequently, the Optical Transport Failure alarm.

Pre-requisite Checks

- Verify redundant link status to ensure continued network connectivity.
- Dispatch a skilled engineer to the Whitefield East POP to facilitate remote hands support.
- Check power supply units and ensure they are functioning within nominal parameters.

Potential Root Causes

Cause	Probability
Degraded or faulty SFP/XFP module	High
Fiber cut or damage	Medium
Rx/Tx power loss due to misaligned or damaged optical connectors	Low

Diagnostic Commands & Expected Outputs

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Command: show opt trans mib, Expected Output: Optical Transport MIB details, including 10G-ER-LD module status and Rx/Tx power levels., Description: This command retrieves optical transport MIB details, providing insight into the 10G-ER-LD module's operational status, including Rx/Tx power levels, and facilitating the identification of any issues within the optical transport system.

Command: show system sfp, Expected Output: SFP/XFP module details, including status, temperature, and voltage parameters., Description: This command displays SFP/XFP module details, including status, temperature, and voltage parameters, allowing the engineer to assess the operational condition of the SFP/XFP modules and identify any potential issues.
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Resolution Steps

Step 1: Step: Replace the degraded or faulty SFP/XFP module, Command: replace sfp, Description: Replace the degraded or faulty SFP/XFP module with a new one to resolve the Optical Transport Failure alarm. Ensure to follow proper insertion and extraction procedures to avoid damaging the optical connectors or the SFP/XFP module itself.

Step 2: Step: Verify the fiber connections and ensure they are secure and not damaged, Command: verify fiber connections, Description: Verify the fiber connections and ensure they are secure and not damaged. Check for any signs of physical damage or misalignment, and correct any issues found to prevent further Optical Transport Failures.

Post-Check Verification

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Command: show opt trans mib, Expected Output: Optical Transport MIB details, including 10G-ER-LD module status and Rx/Tx power levels., Description: Verify that the Optical Transport MIB details have returned to normal, indicating that the issue has been resolved.

Command: show system sfp, Expected Output: SFP/XFP module details, including status, temperature, and voltage parameters., Description: Verify that the SFP/XFP module details are within normal operational parameters, indicating that the replacement or correction of the fiber connections has been successful.
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Roll-back Procedures

- Step: Roll back to the previous SFP/XFP module (if replaced), Command: replace sfp rollback, Description: If the replacement of the SFP/XFP module did not resolve the issue, roll back to the previous module to avoid further network disruptions.

- Step: Re-verify fiber connections, Command: verify fiber connections, Description: Re-verify the fiber connections and ensure they are secure and not damaged to prevent further Optical Transport Failures.